Ex:1 | Implementation of subnetting in cisco packet tracer

Aim:
Implementation of subnetting in cisco packet Tracer simulation

Procedure:

1) Creating Network Topology
   i) open cisco packet Tracer
   ii) select new → Network → Generic to create a blank topology
   iii) Add switches and PC's by selecting them from the bottom left devices panel and dragging them into the WS
   • Connect the devices using appropriate cables.

2) Subnetting:
   Given network: 192.168.10.1/24
   Subnet mask: 255.255.255.240
   Number of subnet: 8
   Usable host address per subnet: 30
   IP address
   Router R1
   G0 10.192.168.10.0/27
   G1 : 192.168.2.1/27

Send Packet

Send Packet

From: computer1

To: computer1

Packet Type: arp_request

Data: computer6

Send Packet

Send Packet

From: computer4

To: computer4

Packet Type: arp_request

Data: computer6

Send Packet

EXP:9

AIM:

Procedu

1)

2)

3)

4)

5)

6)

```
import java.io.*;
import java.net.*;

public class UDPClient {
    private static final String SERVER_ADDRESS = "127.0.0.1";
    private static final int SERVER_PORT = 9876;

    public static void main(String[] args) throws IOException {
        DatagramSocket socket = new DatagramSocket();
        InetAddress address = InetAddress.getByName(SERVER_ADDRESS);

        String message = "Hello, UDP Server!";
        byte[] sendData = message.getBytes();
        DatagramPacket sendPacket = new DatagramPacket(sendData, sendData.length, address, SERVER_PORT);
        socket.send(sendPacket);

        byte[] receiveData = new byte[1024];
        DatagramPacket receivePacket = new DatagramPacket(receiveData, receiveData.length);
        socket.receive(receivePacket);

        String response = new String(receivePacket.getData(), 0, receivePacket.getLength());
        System.out.println("Server response: " + response);

        socket.close();
    }
}
```

```
PS C:\Users\Lisa\Desktop\New folder (2)> java UDPClient
Sent: UDP packet test
UDP Echo: UDP packet test
PS C:\Users\Lisa\Desktop\New folder (2)>
```

Router R2

F010: 192.168.3.1/27

F011 : 192. 168.4.1 ./27

3) Configure Router

enable

configure terminal

interface fast [Start 0/0

ip address <ip address> <subnetmask>

no shut down

exit

ip address <ip address> <sub net mask

no shutdown

exit

Giga bit (Ethernet configuration)

enable

configure terminal

Interface Giga bit Ethernet 0/0.

4) Configure PCs

For each PC right-click → config

1) IP address

2) DNS Server information

6) Testing the network

a) Open CMD on PC

b) PENCI <destination -IP>

c) Ping other PC's to verify connectivity

d) Ping the router from the PC's to verify routes connectivity

Results:

The classless IP subnetting network was successfully implemented in Cisco Packet Tracer using the 192.168.1.0/24 network subnet IP address is.

Connectivity was verified successfully using ping command.

